

■ ■ SnapRedact AI

On-Device PII Redaction & Document Privacy Tool

Overview

SnapRedact AI is an on-device AI-powered application that automatically detects and redacts Personally Identifiable Information (PII) from documents — including text files, PDFs, images, and DOCX files. Built specifically for Snapdragon-powered HP PCs, all processing occurs entirely on-device using the Snapdragon NPU, ensuring that sensitive data never leaves the user's laptop.

Problem Statement

Organizations across sectors — healthcare, finance, education, HR — handle sensitive documents daily. Before these documents can be shared, PII such as names, Aadhaar numbers, PAN numbers, email addresses, and phone numbers must be identified and removed.

Current solutions have critical limitations:

- Cloud-based tools send data to external servers, creating privacy risks
- Manual redaction is slow, expensive, and error-prone
- Most tools don't support India-specific PII (Aadhaar, PAN)

This is particularly critical as India's Digital Personal Data Protection Act (DPDPA 2023) mandates strict handling of personal data.

Solution

SnapRedact AI combines two complementary AI approaches:

- **AI-Based Named Entity Recognition (NER):** A BERT model (dslim/bert-base-NER) running through ONNX Runtime detects context-dependent PII — person names, organizations, and locations — by understanding natural language context.
- **Intelligent Pattern Matching:** Compiled regex patterns detect India-specific structured PII (Aadhaar: 12-digit format, PAN: ABCDE1234F format) and universal identifiers (emails, phone numbers, credit cards, IP addresses, dates of birth).
- **OCR Integration:** Tesseract OCR with image preprocessing extracts text from scanned documents and images, enabling PII detection even in non-digital

documents.

Snapdragon NPU Optimization

- NPU Acceleration: ONNX Runtime leverages Snapdragon NPU for BERT inference
- 100% On-Device: Zero network dependency — works anywhere, even offline
- Low Latency: NPU-accelerated inference for near real-time PII detection
- Model Caching: Model loads once and persists across sessions

Technical Stack

- Python 3.11+ with Streamlit web framework
- Hugging Face Transformers (BERT NER) + ONNX Runtime
- Tesseract OCR + PIL for image processing
- PyMuPDF for PDF handling, python-docx for DOCX support

Key Features

- Detects 11+ types of PII including India-specific identifiers
- Supports text, PDF, DOCX, and image file formats
- OCR for scanned documents with image preprocessing
- Side-by-side original vs. redacted view
- Export as TXT, DOCX, or JSON detection report
- Beautiful, interactive Streamlit dashboard with real-time statistics

Impact

SnapRedact AI demonstrates how on-device AI on Snapdragon-powered PCs can solve a real-world privacy challenge — enabling organizations to comply with data protection regulations (DPDPA 2023, GDPR) while keeping sensitive data completely private and secure.

Author: A V U Pushyami Reddy | pushyamireddy4533@gmail.com
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